HOMEWORK 4

Data Science vs PM2.5

Problem background

Clean air is one of the fundamentals of human health, as air surrounds us everywhere and we breathe it every second. Nevertheless, in the current situation, high technology produces pollution, making the air harder to breathe. PM2.5 is one of the most concerning air pollutants and is becoming increasingly lethal and harmful to human health. Northern Thailand has recurrent dry-season smoke shaped by agricultural and forest burning, regional transport, meteorology, and terrain that can retain pollution. This crucial ongoing situation has led to Thailand's 24-hour PM2.5 standard being set at 37.5 μg/m³ as of 1 June 2023.

Therefore, PM2.5 can penetrate deep into the lungs, enter the bloodstream, and, at the population level, is associated with many diseases, especially respiratory diseases such as asthma and chronic obstructive pulmonary disease (COPD) exacerbations, lower respiratory infections, and lung cancer. A useful warning system should provide preparation time for health communication and services; however, a local project must distinguish established external health evidence from what its own health dataset can demonstrate.

Consequently, the researchers would like to provide the model accuracy for predicting the next-day PM2.5 using multivariable characteristics and explore an association between Pexposure ande and the incidence of respiratory diseases. The researchers proposed that the selected RF improves persistence in later-year evaluations and yields a well-predicted association between PM2.5 and related respiratory disease outcomes.

Approach

In this project, the researcher aims to follow the data science workflow from the problem background as shown in Supplement Figure 1. The researchers would like to acquire data from many sources, including open data source websites, API servers, and web scraping. In this particular project, the researcher provided resources that were already fetched for a frozen path to reproduce what the researcher analyzed. Then, data preparation was required to clean and validate tables to build an analysis-ready table. After that, descriptive analysis was conducted, and graphs were generated from the results. Finally, the researcher created data modeling work, adding context to answer the research question. The script files used are shown in Supplement Figure 2 for both live acquisition and frozen acquisition.

Methodology

Data source exploration

Data were obtained from six datasets using APIs and public open-data sources. OpenAQ provided daily observed PM2.5 measurements by sensor location through a key-authenticated API, while the Open-Meteo Air Quality and Archive APIs supplied daily modeled PM2.5 and weather data. NASA FIRMS VIIRS provided satellite-detected fire hotspot records through a key-authenticated API. DDC/MOPH respiratory records and MOPH facility data were obtained from public open-data sources to represent respiratory health outcomes and healthcare capacity, respectively. The data sources are provided in supplementary Table 1.

Directory preparation

The reproducible workflow consists of seven sequential processes as shown in Supplement Figure 2. Process 0 uses requirements.txt to establish the environment and verify imports. Process 1 runs fetch_data.py to acquire raw source files and generate the fetch manifest, while Process 2 runs prepare_tables.py to create eleven standardized source tables. Process 3 uses prepare_data.py to construct the AT1 and AT2 analysis-ready and model-ready datasets. Process 4 runs check_dataset.py as a feasibility gate, allowing continuation only when FAIL flag appears none. Process 5 executes analysis.py to produce descriptive tables and figures, and Process 6 runs modeling.py to generate AT2 association estimates and AT1 predictions and performance metrics. Finally, once the models are complete, the report is finalized by combining them with the context.

Data Table manipulations

Raw data from each source were first processed separately using `prepare_tables.py`. The purpose of this stage was to ensure the sources were consistent and reliable before deciding how to combine them for analysis. The script checked the schemas, required columns, data types, units, date coverage, row counts, missing values, duplicate records, and key uniqueness.

According to different data sources, dates, places, and identifiers were recorded in different ways. These values were therefore standardized before any merging was performed. For example, UTC timestamps were converted to `Asia/Bangkok` before the analysis date was created, Buddhist-era health years were converted to Gregorian years, and Thai and English province names were mapped to stable province keys. Unmatched province names or invalid dates were reviewed and manipulated to ensure data consistency.

Missing values were also kept separate from confirmed zero values. For example, hotspot counts could be recorded as zero only when the source interval was successfully observed, and no hotspot was detected. Based on data exploration, fire hotspots have missing values due to no sensor installation for certain times or due to different satellite-collected data. Decision-making: if the data could not be retrieved, the value remained missing. Outcome values were not estimated or replaced.

Duplicates were checked according to their meaning. Primary and foreign keys were assigned to each table to give meaning to its rows shown in Supplement table 3. Exact copies of the same record were treated differently from multiple valid observations. For example, several PM2.5 sensors could report on the same day in the same province, while several hotspots could be detected in the same area. These records were not automatically deleted because they represented separate measurements or events. They were retained until further analysis would aggregate data.

At this stage of data preparation, additional source-specific preparation was required. For example, hotspot locations were organized into distance bands (categorical groups) of 0–50 km, 50–100 km, 100–300 km, and 300–500 km—this preserved information about how close each fire detection was to the study province.

Analysis Table maipulation

After the standardized source tables were created, each approved source was converted to the required time and place level before merging. For example, multiple PM2.5 sensor measurements were summarized using the province-day median, weather records were summarized by province and day, and hotspot detections were counted by date and distance band.

Data types were checked again before aggregation and merging. Province and facility identifiers were stored in a consistent format so that the same identifier was not interpreted differently across sources. Dates were parsed as date–time values rather than ordinary text, while PM2.5, weather, hotspot, population, and health variables were converted to numeric values. Values that could not be converted were kept as missing and reported for review instead of being changed to zero.

Duplicate records were examined before creating the analytical tables. Exact duplicate rows were distinguished from multiple valid observations with the same province and date. For example, measurements from different PM2.5 sensors in the same province were valid observations and were combined using the selected aggregation method. After aggregation, the researcher checked that each table contained only one row for its intended time–place key.

Missing values were handled according to their meaning. Missing outcomes were not imputed. A hotspot count was recorded as zero only when the source interval was successfully observed and no hotspot was detected; otherwise, it remained missing. Predictor values that were unavailable also remained missing until the eligibility rules for the final analysis were applied. This prevented unavailable data from being interpreted as no pollution, no fire, or no disease.

Before each merge, the researcher checked that the joining key was unique on the required side of the relationship. This prevented one source row from matching several unintended rows and incorrectly increasing the dataset size. A complete time–place grid was created first, and the source tables were then added using left joins. As a result, dates or provinces without observations remained visible as missing values instead of disappearing from the analytical table.

Time-dependent predictors were created separately within each province after the records had been ordered by date. Lagged variables were calculated before they were joined to the outcome. This ensured that the predictors represented information available before the outcome being studied and prevented future information from entering the analysis.

The final analytical tables were separated only after the data quality, research questions, required variables, and units of analysis had been considered. Therefore, their separation reflected the different analytical purposes and was not imposed during the initial preparation of the source tables.

Data Analysis

The analysis was organized around six analytical sub-questions derived from the study objectives. The first three questions used descriptive analysis to explain data availability and the main patterns of observed PM2.5. The remaining three questions used statistical and predictive modeling to examine forecast performance, warning usefulness, and the association between PM2.5 and respiratory diagnosis-record reporting.

Question 1: *How complete were the PM2.5 observations, and how did observed PM2.5 vary over time and across provinces?*

The percentage of available PM2.5 observations was calculated to assess data completeness. A province-by-year coverage heatmap was used because its color pattern makes provinces and years with limited observations easy to identify. A location map was also created to show the geographic distribution of PM2.5 monitoring stations and whether the study provinces were equally represented. In addition, to examine changes over time, monthly median PM2.5 was plotted as a line graph for each province.

Question 2: *How often did observed PM2.5 exceed 37.5 µg/m³ across provinces and periods?*

Each observed province-day was classified as having a PM2.5 concentration above 37.5 µg/m³. The number and percentage of exceedance days were then calculated by province, calendar month, and year. Heatmaps were used to display exceedance percentages because they allow many province-period combinations to be compared in a single figure.

Question 3: *Which environmental factors were descriptively associated with observed PM2.5?*

The study explored the relationships between observed PM2.5 and fire-hotspot and weather variables. The relationship between same-day Thailand hotspot activity and observed PM2.5 was displayed using a scatterplot. Importantly, hotspot counts were highly uneven; the horizontal axis used log(1 + hotspot count). A binned-median line was added to make the overall pattern easier to see. Moreover, separate scatterplots were used to compare prior-day temperature, relative humidity, precipitation, and wind speed with next-day PM2.5. Pearson and Spearman correlations were displayed in a correlation-coefficient plot, with Spearman correlation used as the main measure. This plot enabled comparison of the direction and strength of several predictor–target relationships in a single figure.

Question 4: *Can next-day PM2.5 be predicted more accurately than the persistence baseline, and does adding hotspot information improve prediction?*

The prediction target was the observed province-day median PM2.5. Predictors included PM2.5 from the previous one to three days, prior-day weather, calendar seasonality, province, and lagged Thailand hotspot counts by distance band. In this study, three prediction approaches were compared: the previous-day persistence baseline, a Random Forest without hotspot predictors, and a Random Forest with hotspot predictors. A grouped bar chart was used to compare the MAE of each approach during validation and testing.

Question 5: *Can the selected model identify high-PM2.5 warning days, and where does it show the largest errors or systematic bias?*

An observed-versus-predicted scatterplot was used to evaluate the selected model on the locked test data. The 1:1 diagonal line represents perfect agreement between observed and predicted PM2.5. Specifically, horizontal and vertical lines at 37.5 μg/m³ divided the graph into correctly warned, missed, false, and correctly identified lower-PM2.5 days. Recall, precision, specificity, and F1 score were calculated to summarize the warning classifications. Also, two province-by-season heatmaps were used. The first showed MAE to identify where the model had the largest errors. The second showed the mean residual, calculated as observed minus predicted PM2.5, to identify systematic bias.

Question 6: *Is higher monthly PM2.5 associated with more respiratory diagnosis records per facility?*

The health analysis used monthly median observed PM2.5 as the exposure and respiratory diagnosis records per active reporting facility as the outcome. The first panel was a province-month scatterplot, using different colors for provinces, different symbols for years, and month labels beside the observations. Then, the second panel displayed the adjusted OLS-HC3 relationship with its 95% confidence interval. The fitted line showed the estimated direction of the association after accounting for province and recurring calendar-month patterns, while the shaded interval showed the uncertainty around that estimate. Spearman correlation and univariable OLS-HC3 were calculated before fitting the adjusted model. The final results included the estimated PM2.5 coefficient, 95% confidence interval, and p-value.

Data Modeling

Two types of analysis were used. A Random Forest model was used to predict next-day PM2.5, while regression models were used to examine the relationship between monthly PM2.5 and respiratory diagnosis records. For PM2.5 prediction, each row represented one province and one day. The target was the observed PM2.5 on the next day. The predictors included: observed PM2.5 from the previous 1–3 days, weather from the previous day, hotspot counts from the previous 1–3 days, province, and time of year. Lagged variables were used because current PM2.5 can be affected by conditions from previous days. They were created separately within each province and ordered by date. This prevented information from the target day or future dates from entering the model. Missing target values were not filled because an estimated target could change the true model performance. Only rows containing all required variables were used. Province was treated as a category using one-hot encoding because province codes do not represent numerical order. Numerical predictors were kept in their original units because Random Forest does not require feature scaling.

The data were divided by time period: data from 2021–2023 were used to train the candidate set to fit the models, and data from 2024 were used to validate the models. Choosing a separate test dataset was needed to provide a final and fair evaluation. Lastly, data in 2025 were used as the test set to evaluate the final selected model. A time-based split was selected because the model was designed to predict future PM2.5. A random split could mix earlier and later observations, allowing the model to learn from future patterns. This could make the performance appear better than it would be in real use.

All data subsets were compared using different prediction approaches: a persistence baseline, a Random Forest without hotspot predictors, and a Random Forest with hotspot predictors. The persistence baseline predicted the next day's PM2.5 using the previous day's observed value. It was selected because PM2.5 often varies gradually from one day to the next. A more complex model should perform better than this simple method, demonstrating that it adds useful information. Random Forest was selected because PM2.5 may have complex relationships with weather, previous PM2.5, and hotspot activity. These relationships may not follow a straight line. Random Forest can handle nonlinear patterns and interactions between predictors without requiring them to be defined in advance. Two Random Forest models were compared to test whether hotspot information improved prediction. The first model used previous PM2.5, weather, province, and time of year. The second model used the same predictors with additional lagged hotspot counts. Keeping the other predictors the same allowed the effect of adding hotspot information to be compared fairly.

In the validation subset, several Random Forest settings were tested by varying the maximum tree depth and the minimum number of observations per leaf. Each model used 300 trees. Tree depth and leaf size were tested because they control model complexity. A model that is too complex may learn noise from the training data, while a model that is too simple may miss important patterns. Model performance was evaluated using MAE for the average prediction error, RMSE for errors that place more weight on big deviations, and R2 for how well predictions captured the observed variation. The model with the lowest validation MAE was selected because MAE directly shows the average prediction error in µg/m³. RMSE was used as a second criterion because it places greater weight on large errors. Errors were also summarized by province and season to identify places or periods where the model performed poorly.

After selection, the final model was retrained on the combined training and validation data. This allowed it to learn from all available data before the test period. It was then evaluated once using the locked test data. Predictions were also classified using the PM2.5 warning level of 37.5 µg/m³. Recall, precision, specificity, and F1 score were used because regression metrics alone do not indicate whether the model correctly identifies warning days.

For the respiratory association model, the health analysis used province-month data. Monthly median PM2.5 was the exposure, and respiratory diagnosis records per active reporting facility were the outcome. Spearman's correlation was used first because it can describe a relationship without assuming linearity. OLS regression was then used to estimate the direction and size of the association. HC3 robust standard errors were used because the variation in respiratory records might not be equal across observations. The adjusted model also controlled for province and calendar month because healthcare reporting and seasonal respiratory patterns could differ across provinces and over time. This model was designed to examine an association. It could not confirm that PM2.5 directly caused changes in respiratory diagnosis records.

Results and discussion

Checkpoint from data exploration

C1 · Time

OpenAQ observations, weather records, and NASA FIRMS hotspot detections were aligned to Thailand local time before daily aggregation. For the hotspot data, date and time (required padded to four digits to be same format) were combined to create the acquisition timestamp in UTC. The timestamp was then converted to the Asia/Bangkok time zone, and the Thailand-local calendar date was used as “analysis_date”. The original date and time fields and the derived UTC and local timestamps were retained for auditing. Daily hotspot counts and their lagged predictors were subsequently calculated using the Thailand-local date so that they referred to the same day definition as the PM2.5 target and weather predictors.

C2 · Join handling

Before each merge, the joining key was checked to ensure that it was unique at the required time and place level. The complete province-day grid contained 13,656 rows, and left joins were used so that dates or provinces without observations remained in the dataset. After joining, 10,047 rows had an observed PM2.5 target, and 8,222 rows had all required target and predictor values for modeling. For the monthly analysis, the complete grid contained 288 possible province-months, of which 35 met the analysis eligibility rules. Row counts were checked before and after every merge to confirm that no unintended many-to-many join increased the dataset size. Therefore, the reduction in usable rows resulted from missing observations and eligibility rules rather than rows being accidentally lost during merging.

C3 · Missing values

The percentage of missing values was calculated for every column before analysis. Observed PM2.5 was unavailable for 3,609 of the 13,656 province-days, equal to 26.43%. The PM2.5 target was never imputed, and rows without the required target or predictor values were excluded only when the model-ready dataset was created. A confirmed zero hotspot count on a successfully retrieved date was kept as zero, whereas a date with unavailable FIRMS data remained missing. Columns with 0% missing values were usually constructed fields, such as province keys, planned grid dates, or derived calendar variables.

C4 · Comparing places

Province-level data were checked before spatial comparison to confirm that the eight provinces contained distinct observations rather than repeated copies of the same series. OpenAQ sensor coordinates were used to confirm which monitoring stations were located within each target province and to show the actual monitoring coverage. All quality-passing sensors in a province were retained, and measurements from multiple sensors on the same day were summarized using the province-day median. Therefore, the resulting PM2.5 value represented the available provincial monitoring network rather than complete exposure across the entire province. Hospital and health-facility locations were standardized using the official province key and were used to identify the number of active reporting facilities within each province-month. Individual hospitals were not matched to the nearest PM2.5 sensor, because the respiratory outcome was available only as a province-month total and did not represent patient-level exposure at each hospital. Province boundaries were used as the common geographic reference for checking coordinates, assigning location-based records to provinces, and identifying unmatched or incorrectly located records before aggregation. The boundaries were not used to claim that monitoring coverage was uniform within each province. Consequently, province comparisons were interpreted as conditional on the available sensor and health-facility coverage rather than as equally measured population-wide differences.

Data analysis results

According to the analysis, the descriptive analysis shows the data coverage each province with mean with standard deviation, median, range and numbers of day that exceed standard PM2.5 with percentage shown in Table 1. From the tables, most coverage province is Chiang Mai with 83.6 %, while Phayao only has 54.6 % coverage. In every province, the mean was higher than the median, and the maximum was far above the upper quartile. Nan had the highest median PM2.5 (21.4 µg/m³) and the highest percentage of observed days above 37.5 µg/m³ (23.9%). Mae Hong Son had the highest mean (28.0 µg/m³) and the largest standard deviation (40.5 µg/m³). Chiang Rai had the highest single province-day value (423.5 µg/m³). The low median but high mean in Mae Hong Son shows that a limited number of severe episodes increased its average.

Table 1 *Distribution and completeness of daily province-level PM2.5 observations, 2021-2025*

Province	Observed days (n)	Coverage (%)	Mean ± SD, µg/m³	Median (Q1-Q3), µg/m³	Range, µg/m³	Days >37.5 µg/m³ n (%)
Chiang Mai	1,427	83.6	25.5 ± 26.0	17.2 (9.7-32.0)	1.1-227.0	259 (18.1)
Lamphun	1,311	76.8	26.1 ± 21.8	20.3 (9.9-35.3)	2.3-176.0	290 (22.1)
Lampang	1,431	83.8	23.2 ± 21.9	15.5 (8.4-31.5)	0.6-176.0	255 (17.8)
Phrae	1,068	62.6	24.3 ± 19.6	16.8 (9.9-34.6)	0.0-114.2	229 (21.4)
Nan	1,222	71.6	28.7 ± 24.8	21.4 (12.1-36.0)	0.5-228.0	292 (23.9)
Phayao	932	54.6	26.7 ± 26.8	16.8 (9.3-34.0)	5.2-236.0	201 (21.6)
Chiang Rai	1,236	72.4	27.7 ± 35.0	18.3 (9.6-32.5)	0.8-423.5	225 (18.2)
Mae Hong Son	1,420	83.2	28.0 ± 40.5	15.0 (6.0-30.2)	0.9-357.0	273 (19.2)

Question 1: *How complete were the PM2.5 observations, and how did observed PM2.5 vary over time and across provinces?*

Figure 1 shows how complete the PM2.5 data were, where the monitoring locations were placed, and how PM2.5 changed over time. The study included 13,656 possible province-days. PM2.5 was observed on 10,047 province-days (73.6%), while 3,609 province-days (26.4%) had no observed value. Coverage differed by province. It was lowest in Phayao (54.6%) and highest in Lampang (83.8%). Coverage also differed by year. In 2023, is the lowest year coverage was especially low in Phayao (20.0%), Nan (23.0%), Chiang Rai (24.1%), Phrae (24.7%), and Lamphun (38.6%), however improvement is occuring in 2024 and 2025 for most provinces, but it remained lower in Phayao and Phrae. The monthly lines show a clear seasonal pattern. PM2.5 rose during about February to April and fell during the wetter months. The strongest peak occurred in March-April 2023, especially in Mae Hong Son and Chiang Rai.

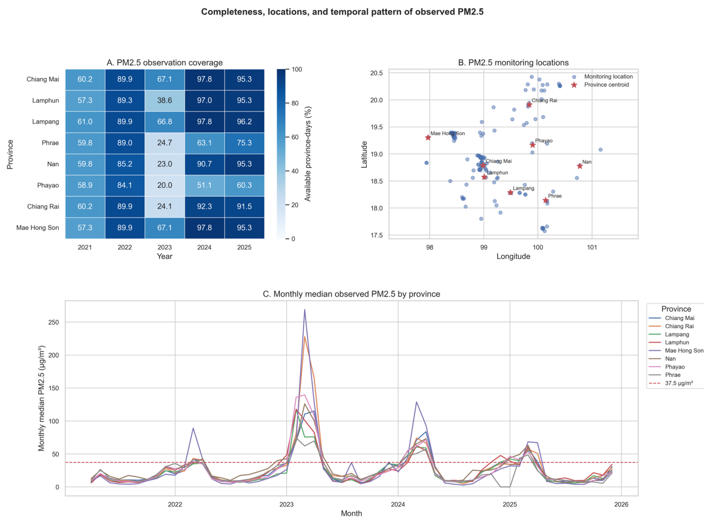

Figure 1 *Completeness, monitoring locations, and monthly patterns of observed PM2.5.*

Question 2: *How often did observed PM2.5 exceed 37.5 µg/m³ across provinces and periods?*

Figure 2 shows that high-PM2.5 days were concentrated in the late dry season especially in March in every province. When all available years and provinces were combined, the median PM2.5 was 56.5 µg/m³ in March and 53.0 µg/m³ in April. The percentage of observed province-days above 37.5 µg/m³ was 82.6% in March and 70.4% in April. It was 49.3% in February, 26.3% in January, and 14.1% in May. From June to November, the percentage was low, ranging from 0.0% to 2.9%, before increasing to 17.8% in December. The same seasonal pattern appeared in all provinces. In March, the percentage above 37.5 µg/m³ ranged from 74% in Phrae to 93% in Mae Hong Son. In April, it ranged from 63% in Lamphun to 81% in Mae Hong Son. Six provinces had their highest yearly percentage in 2023. Nan had the highest yearly value at 61.9%. Chiang Mai and Lampang reached their highest values in 2024, at 30.2% and 31.8%. By 2025, the yearly percentages had fallen to 16.7%-23.2%.

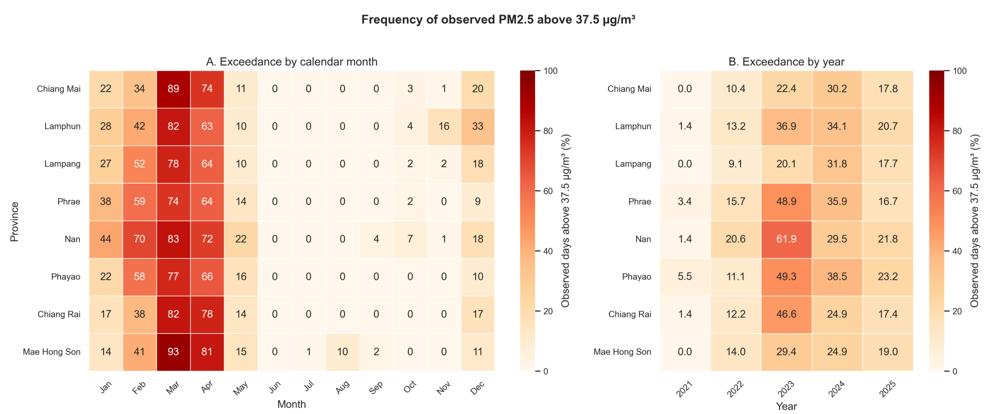

Figure 2 *Percentage of observed province-days with PM2.5 above 37.5 µg/m³ by month and year*

Question 3: *Which environmental factors were descriptively associated with observed PM2.5?*

Figure 3A shows that PM2.5 generally increased when the number of Thailand fire hotspots increased. The relationship was positive in all 40 province-year groups. The Spearman correlation ranged from 0.557 to 0.926. This was a consistent descriptive pattern, but the wide spread of the points shows that hotspot count alone did not explain every high-PM2.5 day. Higher PM2.5 was also more common under dry conditions. Prior-day relative humidity and rainfall had negative Spearman correlations with next-day PM2.5

(rho = -0.613 and -0.687). Temperature had a weak negative rank correlation (rho = -0.240), but its Pearson correlation was close to zero (r = 0.008; FDR q = 0.649). Wind speed had only a weak positive correlation (rho = 0.118). These simple correlations can be affected by season and by relationships among weather variables. They do not show an independent cause, model importance, or prediction accuracy.

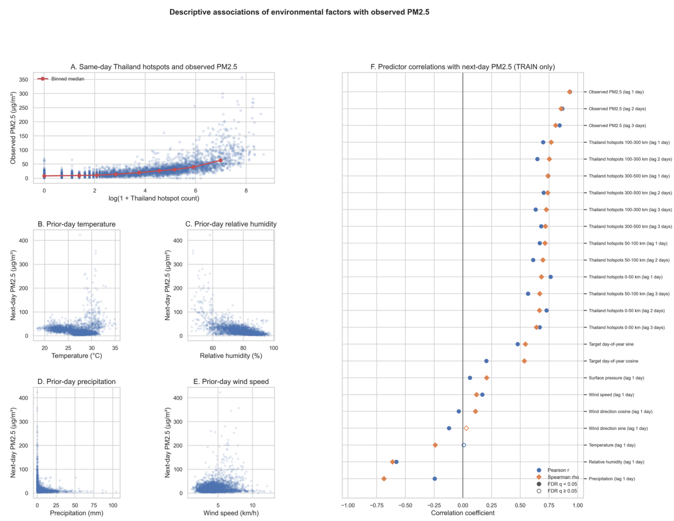

Figure 3 *Descriptive relationships between environmental factors and observed PM2.5*

Question 4: *Can next-day PM2.5 be predicted more accurately than the persistence baseline, and does adding hotspot information improve prediction?*

Table 2 *Performance of candidate one-day-ahead PM2.5 models in the 2024 validation and locked 2025 test periods*

Model	Validation MAE	Test MAE	Test RMSE	Test R^2
Persistence	5.913	4.613	8.209	0.871
RF without hotspots	5.951	4.458	7.755	0.885
RF with hotspots (selected)	**5.822**	**4.369**	**7.535**	**0.891**

The model-ready dataset contained 8,222 province-day rows. The split followed time: 3,657 rows from 2021-2023 were used for training, 2,153 rows from 2024 were used for model selection, and 2,412 rows from 2025 were kept as a locked test set. The test set was opened only after the model was selected. Rows from the same date remained in the same split. This design tests performance on a later year and reduces the risk of information leaking across time. The MAE amonge different validation scenarios were shown in Figure 4 and table 2

In the 2024 validation test was performed, it had the lowest MAE at 5.822 μg/m³, compared with 5.913 for persistence and 5.951 for the RF without hotspots. So, the RF with hotspot was the condition use to test the data. On the locked 2025 test set, the selected model had an MAE of 4.369 μg/m³. In simple terms, its prediction differed from the observed value by about 4.37 μg/m³ on average. It reduced MAE by 5.28% compared with persistence and by 1.98% compared with the RF without hotspots. Its RMSE was 7.535 μg/m³ and its R^2 was 0.891. The RMSE was clearly higher than the MAE, which shows that some days still had much larger errors. The R^2 means that the model captured much of the variation in the test data; it should not be read as 89.1% accuracy.

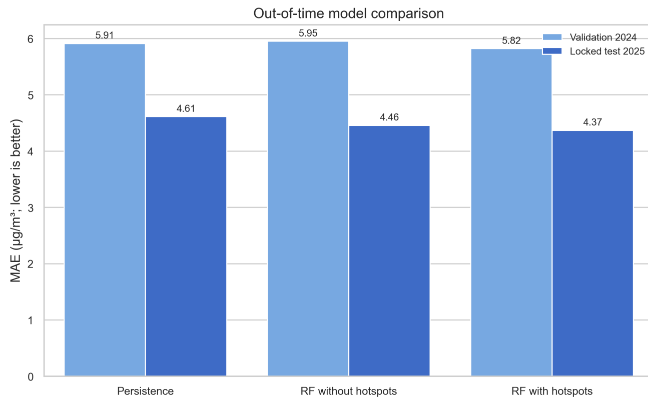

Figure 4 *Out-of-time comparison of validation and locked-test MAE for the three candidate models*

Question 5: *Can the selected model identify high-PM2.5 warning days, and where does it show the largest errors or systematic bias?*

Table 3 *Warning classification of the selected RF with hotspots at the 37.5 µg/m³ threshold on the test set*

Observed PM2.5 status	Predicted ≥37.5 µg/m³	Predicted <37.5 µg/m³	Total
Observed ≥37.5 µg/m³	391	54	445
Observed <37.5 µg/m³	67	1,900	1,967
Total	**458**	**1,954**	**2,412**

The model correctly identified 391 of 445 observed warning days, giving a recall of 87.9%. It missed 54 warning days and produced 67 false warnings. Precision was 85.4%, so most model warnings were correct, while specificity was 96.6%, showing strong performance on days below the threshold. These values support use as an early preparation signal, but the 54 missed warning days are too important to allow the model to issue or cancel warnings on its own.

Low and moderate PM2.5 predictions were close to the ideal 1:1 line as shown in Figure 5. The points became more widely spread when observed PM2.5 was high. This pattern means that the model was generally more reliable on ordinary days than during severe pollution episodes. The threshold quadrants also show both false warnings and missed warnings, so the regression result must be read together with warning-classification performance. Season also affects the precision of prediction , hot season was the most difficult period in every province. Hot-season MAE ranged from 5.2 µg/m³ in Phrae to 12.8 µg/m³ in Mae Hong Son. Errors were lower in the cool season (3.3-4.3 µg/m³) and rainy season (1.2-3.1 µg/m³). Consequently, the mean residual as observed minus predicted PM2.5. A positive value means underprediction, while a negative value means overprediction. During the hot season, the model underpredicted on average in Mae Hong Son (+1.4 µg/m³) and Chiang Rai (+0.4 µg/m³), but overpredicted in the other provinces, especially Lampang (-3.5 µg/m³) and Chiang Mai (-3.0 µg/m³). For severe days with observed PM2.5 ≥75 µg/m³, MAE increased to 16.84 µg/m³ and the mean residual was +6.15 µg/m³. The model therefore tended to understate the most severe events .

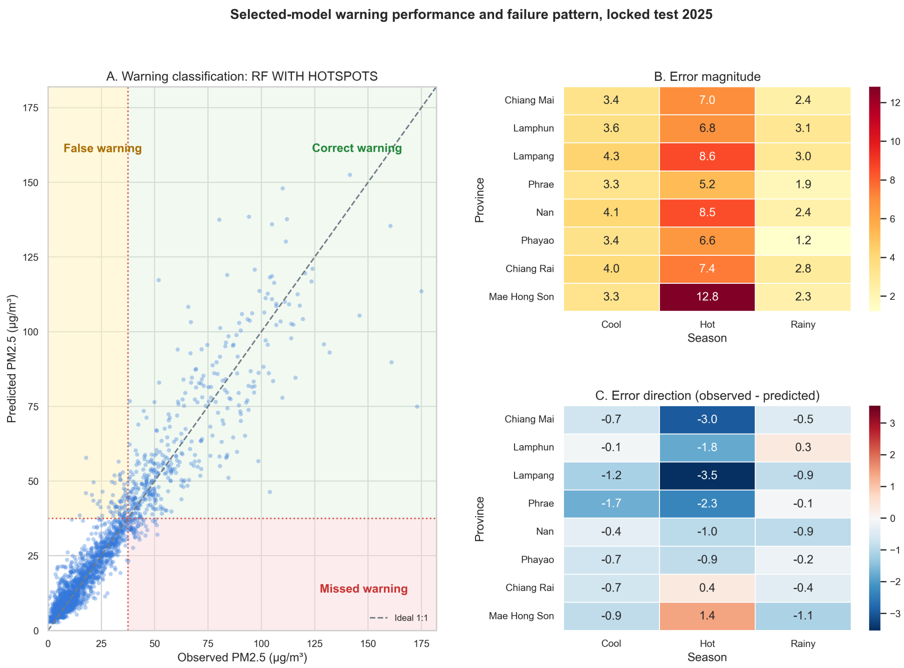

Figure 5 *Warning classification, error magnitude, and error direction for the selected model on the test set*

Question 6: *Is higher monthly PM2.5 associated with more respiratory diagnosis records per facility?*

Table 4 *Association between monthly median observed PM2.5 and respiratory diagnosis-record reporting*

Analysis	Estimate	Robust SE	95% CI	p-value	R^2	Adjustment
Spearman correlation	ρ = -0.655	—	—	<0.001	—	None
Unadjusted OLS-HC3	β = -46.99	8.62	-64.52 to -29.46	<0.001	0.309	None
Adjusted OLS-HC3	**β = +12.98**	**21.11**	**-30.59 to +56.55**	**0.544**	**0.741**	**Province fixed effects; cyclic calendar month**

The study shows differences in available capacity among provinces. The median number of opened beds ranged from 543 in Mae Hong Son to 5,778 in Chiang Mai. Opened beds per 1,000 population ranged from 1.88 in Mae Hong Son to 3.21 in Chiang Mai. Figure 6A shows the 35 model-eligible province-months. The points form clear groups by province and reporting period. Several low-PM2.5 months had high numbers of records per active facility, while some high-PM2.5 months had lower reported values. This produced a strong negative crude relationship. Before adjustment, higher PM2.5 was associated with lower respiratory reporting. Spearman's ρ was -0.655 ($p<0.001$), and the unadjusted OLS coefficient was -46.99 records per active facility for each 1 µg/m³ higher monthly median PM2.5 (95% CI -64.52 to -29.46; $p<0.001$). After adjustment for province and cyclic calendar month, the coefficient changed direction to +12.98 records per active facility for each 1 µg/m³ higher monthly median PM2.5. However, the 95% confidence interval was wide (-30.59 to +56.55), crossed zero, and the p-value was 0.544. Figure 6B shows the same result: the adjusted line slopes upward, but the shaded 95% confidence interval is wide. The data are therefore compatible with a decrease, little change, or an increase in reporting. The change from a strong crude negative association to a positive adjusted estimate shows that province and seasonal/reporting structure strongly affected the observed relationship. It also shows why the crude correlation cannot answer the research question by itself. At the same time, the adjusted estimate is unstable because only 35 observations were available for a model that also included province and calendar-month terms.

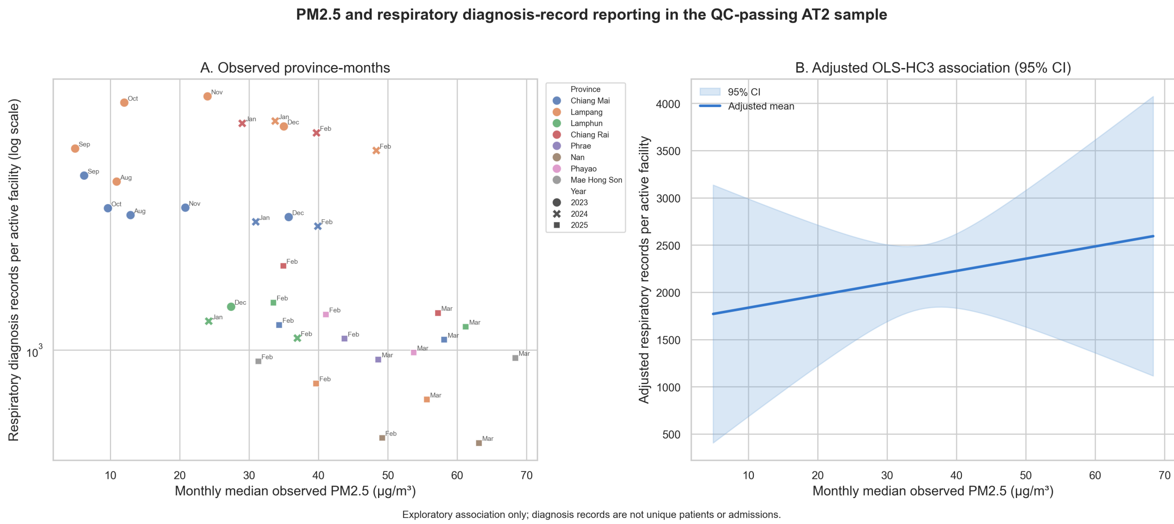

Figure 6. *Observed province-month pattern and adjusted OLS-HC3 association between monthly median PM2.5 and respiratory diagnosis-record reporting per active facility.*

Checkpoint after modeling

C5 · Model versus baseline

The selected random forest model with hotspot information achieved a mean absolute error (MAE) of 4.369 µg/m³ on the locked test set, compared with 4.613 µg/m³ for the persistence model, representing a 5.28% improvement. However, four-fold grouped temporal cross-validation showed that the random forest model achieved a higher average MAE than the persistence model (5.244 versus 4.995 µg/m³) and performed better in only one of four folds. Therefore, the model showed useful performance for the 2025 test period, but its advantage was not consistent across different time periods. The results support its use as supplementary evidence for preparedness and planning rather than as an independent warning system.

C6 · Ground truth

The selected random forest model with hotspot information produced predictions that were reasonably close to the observed values, with a test MAE of 4.369 µg/m³. At the PM2.5 threshold of 37.5 µg/m³, the model gives a recall of 87.9%, precision of 85.4%, specificity of 96.6%, and F1 score of 86.6%. The predictions can therefore be used as an early preparedness signal, but they should not replace current measurements from local monitoring stations. The selected model already included lagged Thailand hotspot counts together with previous PM2.5, weather, and calendar information. In practice, when the model predicts a PM2.5 exceedance, provincial health authorities and hospitals can confirm the situation using local sensor readings and prepare short-term surge capacity, including staff, triage areas, medicines, protective equipment, and respiratory-care resources.

Limitations

This study has several limitations. First, it used a retrospective observational design based on secondary data which were incomplete and unevenly distributed. The results can describe patterns, evaluate prediction performance, and estimate associations, but they cannot prove that fire hotspots caused a specific PM2.5 episode or that PM2.5 caused respiratory disease records to increase. Missing values also appeared that might also have affected comparisons among provinces and years. Additionally, the environmental predictors did not fully represent all sources and movement of air pollution. Fire hotspots were counted detections rather than the amount of smoke emitted. The main hotspot variables covered Thailand and did not fully measure cross-border smoke. Weather data were obtained from gridded estimates rather than measurements from every monitoring station.

The PM2.5 prediction model was evaluated internally using a chronological split rather than external data. In addition, rolling time-series validation did not show consistent improvement over persistence. Model performance may therefore change in future years, particularly if pollution sources, climate, monitoring coverage, or data systems change. Moreover, health analysis had a small and uneven sample. The outcome was the number of respiratory diagnosis-code records per active reporting facility. It was not the number of unique patients, visits, admissions, new cases, or population incidence. Changes in facility participation, coding practice, and reporting completeness may have affected the outcome. opened-bed data provided only general capacity context. They did not show daily bed availability, occupancy, respiratory-bed use, staffing, oxygen availability, or demand during smoke episodes. The study therefore cannot determine whether a province had a bed shortage or how many additional beds would be required.

Recommendations

Provincial public health offices, hospitals, and local governments should begin seasonal preparation before February because the highest PM2.5 burden occurred from February to April. March was the main high-risk month, when 82.6% of observed province-days exceeded 37.5 µg/m³, followed by 70.4% in April. This information can support decisions about the timing of public communication, review of clean-air arrangements, staff readiness, respiratory medicine and equipment checks, and closer surveillance.. Decisions about staffing, respiratory clinics, medicines, oxygen supplies, referral arrangements, or beds should be based on real-time outpatient and emergency demand, admissions, bed occupancy, staffing, and local clinical judgment..

Conclusion

This study found clear seasonal and provincial differences in PM2.5 across Northern Thailand. The selected model produced predictions that generally followed observed PM2.5 patterns and included hotspot, weather, and previous pollution information. However, the predictions should be confirmed with current monitoring-station data before local action. The health analysis did not provide clear evidence that higher PM2.5 directly increased respiratory service demand. Therefore, the model can support early preparation of staff, medicines, equipment, and services, while decisions to expand hospital capacity require more complete healthcare-utilization data.

AI disclosure

Where you used it	What you asked for	What you changed or rejected
Getting raw data	Search for relevant raw-data sources, organize the files, and help write the data-fetching code.	I checked the suggested sources and retained only those that were relevant, accessible, and suitable for the study.
Cleaning data **Manipulating data**	Calculate data coverage and identify duplicate records, missing values, and possible data-quality problems. Identify possible primary and foreign keys and suggest how the datasets could be merged.	I reviewed the results and decided how missing values, duplicates, and incomplete coverage should be handled. I also verified the keys and selected the merge key
Analyzing data **Data model** **Visualizing data**	Help write code for descriptive analysis and statistical modeling. Suggest figure layouts, labels, colors, legends, and titles.	I checked the code, variables, model outputs, and research relevance and selected the clearest layouts and revised figures
Code development **Code review**	Explain required packages, helper functions, and how to run the main functions. Review the plotting and analysis code and identify possible errors or layout problems.	I adapted the code to the actual file structure and computing environment to test the code and corrected by revised code .

Submission checklist (copy the following to your report)

[✓] Repository is public, or the instructor has been added as a collaborator

[✓] README.md explains how to run everything, in order

[✓] requirements.txt lists the packages actually used

[✓] Cloned into a fresh folder and re-ran everything; it worked

[✓] data/raw/ contains exactly what the API returned

[✓] All figures exist as separate files in outputs/figures/

[✓] Model metrics saved in outputs/results/

[✓] Report follows the required structure and is a PDF

[✓] All six checkpoints from Section 7 answered in the report

[✓] Baseline score reported alongside model score

[✓] Recommendation names a specific audience

[✓] AI disclosure table completed

[✓] I can explain every line of code I submitted

Student signature____Solos Lymkhanakhom____Date ___5/9/2026____